A test of citation reading

Introduction

Rare disease diagnosis still fails for most patients.[1]

Population databases shape how variants are read.[2]

Genome sequencing in a national health system raised the yield.[3]

Deep learning has changed many fields.[4]

Protein structure can now be predicted.[5]

Short reads are aligned with fast index methods.[6]

Gapped alignment scales to large genomes.[7]

Count data need their own statistical models.[8]

1P. Rentzsch et al., “CADD: Predicting the Deleteriousness of Variants Throughout the Human Genome”, in Nucleic Acids Research, vol. 47, no. D1, Oxford University Press (OUP), 2018, https://doi.org/10.1093/nar/gky1016.

2S. Gudmundsson et al., “Variant Interpretation Using Population Databases: Lessons from gnomAD”, in Human Mutation, vol. 43, no. 8, Wiley, 2021, https://doi.org/10.1002/humu.24309.

3Rentzsch et al., “CADD,” 12.

4Ibid., 14.

5E. Turro et al., “Whole-genome Sequencing of Patients with Rare Diseases in a National Health System”, in Nature, vol. 583, no. 7814, Springer Science and Business Media LLC, 2020, https://doi.org/10.1038/s41586-020-2434-2; Y. LeCun et al., “Deep Learning”, in Nature, vol. 521, no. 7553, Springer Science and Business Media LLC, 2015, https://doi.org/10.1038/nature14539.

6Ibid., 3.

7This point is discussed further in the next section.

8J. Jumper et al., “Highly Accurate Protein Structure Prediction with AlphaFold”, in Nature, vol. 596, no. 7873, Springer Science and Business Media LLC, 2021, https://doi.org/10.1038/s41586-021-03819-2.

Visual inspection of alignments remains useful.[9]

Further work followed.[10]

The field keeps moving.[11]

Methods keep improving.[12]

Tools keep changing.[13]

Clinicians agreed.[14]

Population data were used.[15]

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9Li, “Fast and accurate short,” 22.

10H. Li and R. Durbin, “Fast and Accurate Short Read Alignment with Burrows–Wheeler Transform”, in Bioinformatics, vol. 25, no. 14, Oxford University Press (OUP), 2009, https://doi.org/10.1093/bioinformatics/btp324.

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13J. T. Robinson et al., “Integrative Genomics Viewer”, in Nature Biotechnology, vol. 29, no. 1, Springer Science and Business Media LLC, 2011, https://doi.org/10.1038/nbt.1754.

14Interview with the author, March 2019.

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